Data visualization using R

Introduction to ggplot2

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Overview

1. Introduction to ggplot2
2. General template of ggplot2
3. Components of ggplot2

Introducing ggplot2: The Grammar of Graphics

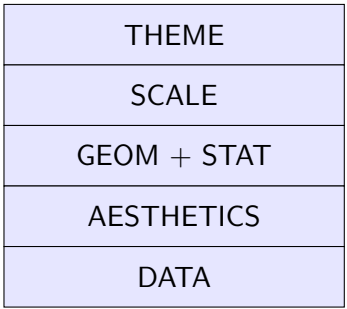
What is ggplot2?

- A system for declaratively creating graphics in R
- Based on the **Grammar of Graphics**
- Builds plots layer by layer

Core Idea:

$$\textbf{Data} \rightarrow \textbf{Map} \rightarrow \textbf{Compute} \rightarrow \textbf{Draw}$$

Plots are constructed from components rather than drawn manually.



Built from bottom to top

General Template of ggplot2

```
ggplot(data = DATAFRAME, aes(x = X_VARIABLE, y = Y_VARIABLE)) +
  geom_GEOMETRY(mapping = aes(),
                stat = "STATISTIC",
                position = "POSITION") +
  labs(title = "Title",
       x = "X label",
       y = "Y label") +
  scale_*() +
  theme_minimal()
```

Component 1: DATA

Provides the dataset (data frame).

```
ggplot(data = mpg)
```

Defines the plotting canvas only.

Component 2: Aesthetics (aes)

Maps variables to visual properties.

- x-axis
- y-axis
- color
- fill

```
ggplot(mpg, aes(x = displ, y = hwy))
```

Still nothing is drawn yet.

Component 3: STAT (Statistical Transformation)

Stat computes before drawing.
Examples:

- Bar plot $\rightarrow$ count observations
- Histogram $\rightarrow$ bin data
- Smooth line $\rightarrow$ regression calculation

Stat calculates, Geom draws.

Component 4: GEOM (Geometry)

Determines what is drawn.

```
ggplot(mpg, aes(displ, hwy)) +
  geom_point()
```

Now the plot appears.

Component 5: Position Adjustment

Controls arrangement of elements.

```
geom_bar(position = "dodge")
```

- stack (default)
- dodge (side-by-side)
- jitter (reduces overlap)

Component 6: Scales

Controls axis limits and visual mappings.

```
scale_y_continuous(limits = c(0, 50))
```

Scales transform data values into visual values.

Component 7: Theme

Controls non-data appearance.

`theme_minimal()`

- Background
- Gridlines
- Fonts

Putting It All Together

```
ggplot(mpg, aes(x = displ, y = hwy, color = drv)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm") +
  theme_minimal()
```

Layer	Role
Data	mpg dataset
Aesthetics	displ, hwy, drv
Stat	Regression (lm)
Geom	Points + smooth line
Theme	Minimal background

Final Mental Model

$$\text{Data} \rightarrow \text{Map} \rightarrow \text{Compute} \rightarrow \text{Draw} \rightarrow \text{Style}$$

This is the grammar of graphics.